

PU-I Biology · Chapter 8

Cell: The Unit of Life

Handwritten Notes — Narendra Sir · BAIO Learning System

1 Cell & Cell Theory

Def ▶ Cell = the fundamental structural & functional unit of all living organisms.

- **Anton von Leeuwenhoek** — first saw & described a live cell · **Robert Brown** — discovered the nucleus
- **Robert Hooke** (1665) — coined the term "cell" (cork)
- **Schleiden** (1838, plants) & **Schwann** (1839, animals) → cell theory · Schwann: cell wall unique to plants
- **Rudolf Virchow** (1855): *Omnis cellula-e cellula* — cells arise from pre-existing cells

★ exam

Modern cell theory

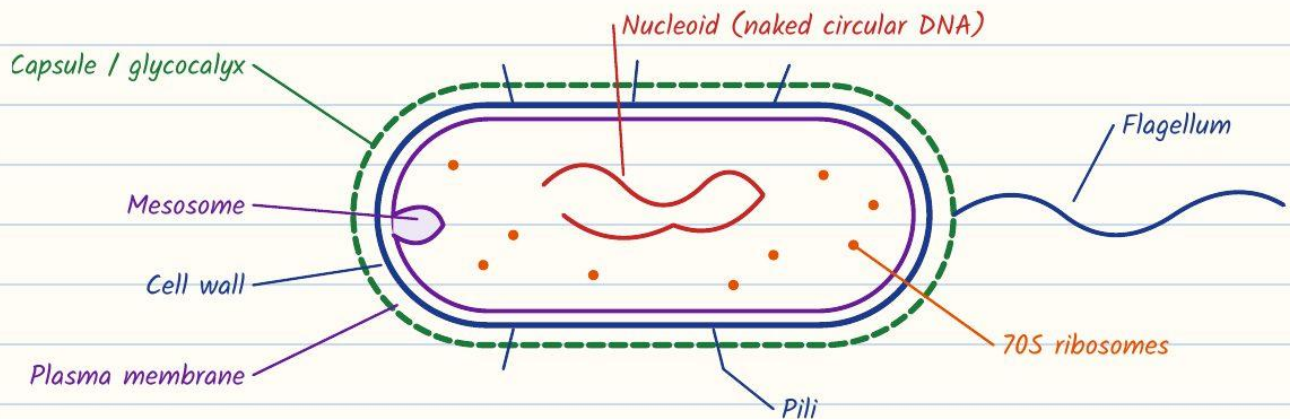
1. All living organisms are composed of cells and products of cells
2. All cells arise from pre-existing cells

Record	Cell
Smallest	Mycoplasma (~0.3 μm)
Largest isolated single cell	egg of ostrich
Longest	nerve cell
RBC (human)	~7.0 μm diameter

- Every cell: **plasma membrane** · **cytoplasm** (main arena of activities) · **nucleus** / **nucleoid**
- Membrane-bound organelles (ER, Golgi, lysosomes, mitochondria, plastids...) only in **eukaryotes** · **ribosomes** (non-membranous) in all cells

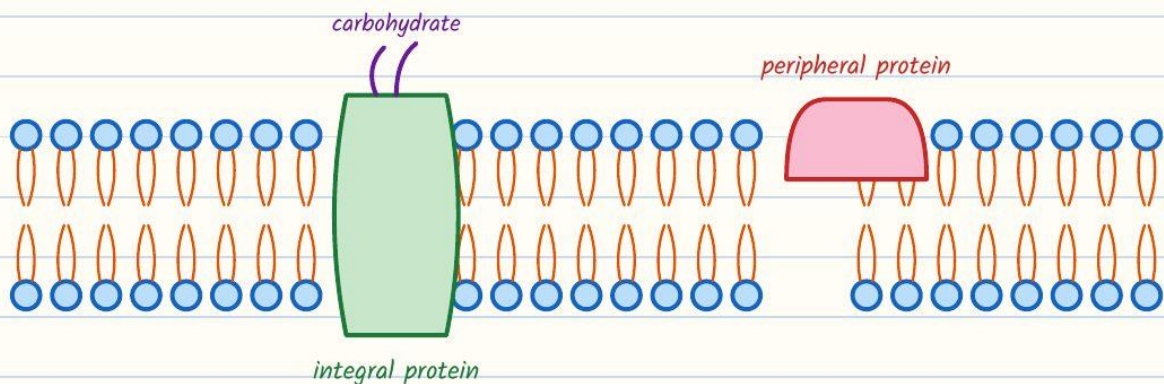
2 Prokaryotic Cells

Bacteria, blue-green algae (cyanobacteria), **mycoplasma** and **PPL0** (pleuro-pneumonia-like organisms).



- No well-defined nucleus — genetic material = naked, circular DNA (**nucleoid**) · extra DNA = **plasmids** (antibiotic resistance)
- **Cell envelope** (3 layers): **glycocalyx** (slime layer — loose, or capsule — thick) · cell wall · plasma membrane ★ **exam**
- Gram-positive (take up the stain) vs Gram-negative (don't)
- **Mesosomes** — infoldings of plasma membrane → cell wall formation, DNA replication & distribution, respiration, secretion · **chromatophores** in cyanobacteria
- Flagellum = filament + hook + basal body · **pili** & **fimbriae** = attachment, not motility
- **70S ribosomes** (50S + 30S) · **polysome** = many ribosomes on one mRNA · inclusion bodies (phosphate, cyanophycin, glycogen granules, gas vacuoles)

3 Cell Membrane & Cell Wall



polar heads (hydrophilic) outside · tails (hydrophobic) inside

Fluid mosaic model — Singer & Nicolson (1972)

- **Lipid bilayer** — mainly **phosphoglycerides**; polar heads outward, hydrophobic tails inward · also cholesterol
- **Human RBC membrane**: ~52% protein, 40% lipid · proteins: **integral** (buried) or **peripheral** (surface)
- **Fluid mosaic model**: quasi-fluid lipids allow lateral movement of proteins · fluidity → cell growth, junctions, secretion, endocytosis, division
- **Passive transport** — along gradient (diffusion, osmosis = water) · **Active transport** — against gradient, uses ATP (Na^+/K^+ pump) ★ exam

Cell wall (non-living, rigid; plants & fungi) — algae: cellulose, galactans, mannans, CaCO_3 · plants: cellulose, hemicellulose, pectins, proteins

Primary wall (young, growing) → **secondary wall** (inner side) · **middle lamella** = calcium pectate, glues cells · **plasmodesmata** connect cytoplasm of neighbours

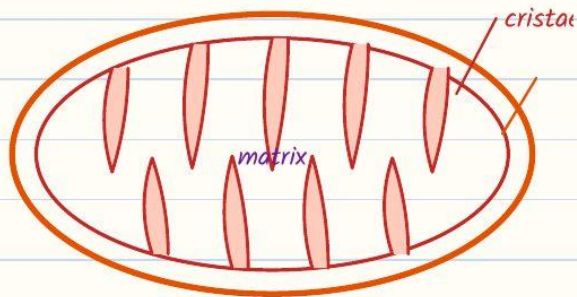
4 Endomembrane System

Def ► ER + Golgi complex + lysosomes + vacuoles — functions are coordinated. (Mitochondria, chloroplasts, peroxisomes are **not** included.)

- **RER** — ribosomes on surface → protein synthesis & secretion · **SER** — lipid / steroid synthesis (animals)
- **Golgi apparatus** (Camillo Golgi, 1898): flat stacked **cisternae** · **cis** (forming) face receives from ER, **trans** (maturing) face releases · packaging & secretion · site of **glycoproteins & glycolipids** formation ★ exam
- **Lysosomes** — from Golgi · **hydrolytic enzymes** (lipases, proteases, carbohydrases) active at **acidic pH**
- **Vacuole** — bound by **tonoplast**; up to 90% of plant cell volume · contractile vacuole (Amoeba: excretion) · food vacuole (protists)

5 Ribosomes & Mitochondria

- **Ribosomes** (Palade, 1953) — RNA + protein, **no membrane** · eukaryotic **80S** (60S + 40S) · prokaryotic **70S** (50S + 30S) · S = Svedberg unit (sedimentation coefficient)



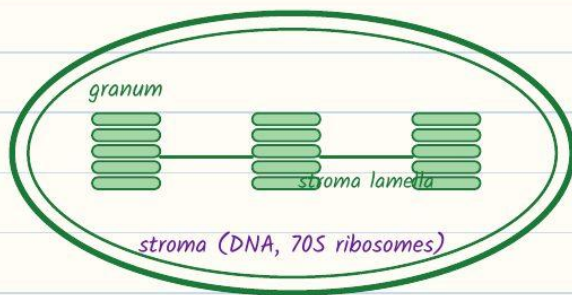
Mitochondrion

- "Powerhouse" — site of **aerobic respiration**, ATP
- Double membrane · inner folds = **cristae** (↑ surface)
- Matrix: single circular DNA, a few RNA, **70S ribosomes**
- Divide by **fission** → semi-autonomous

★ exam

6 Plastids

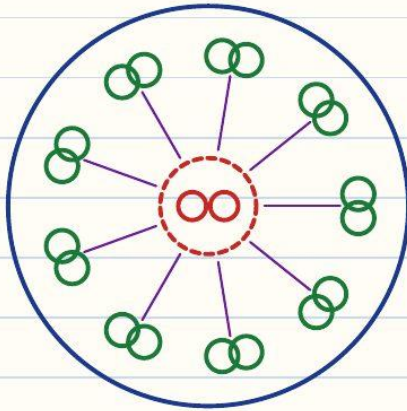
- In all plant cells & euglenoids · **chloroplasts** (chlorophyll, carotenoids) · **chromoplasts** (fat-soluble carotenoids — yellow, orange, red) · **leucoplasts** (colourless, storage)
- Leucoplasts: **amyloplasts** (starch — potato) · **elaioplasts** (oils, fats) · **aleuroplasts** (proteins)



Chloroplast

- Lens-shaped, double membrane (inner less permeable)
- **Thylakoids** stacked = **grana** · joined by **stroma lamellae**
- Stroma: enzymes for carbohydrate & protein synthesis, small circular DNA, **70S ribosomes**
- Chlorophyll in thylakoids

7 Cytoskeleton, Cilia, Flagella & Centrioles

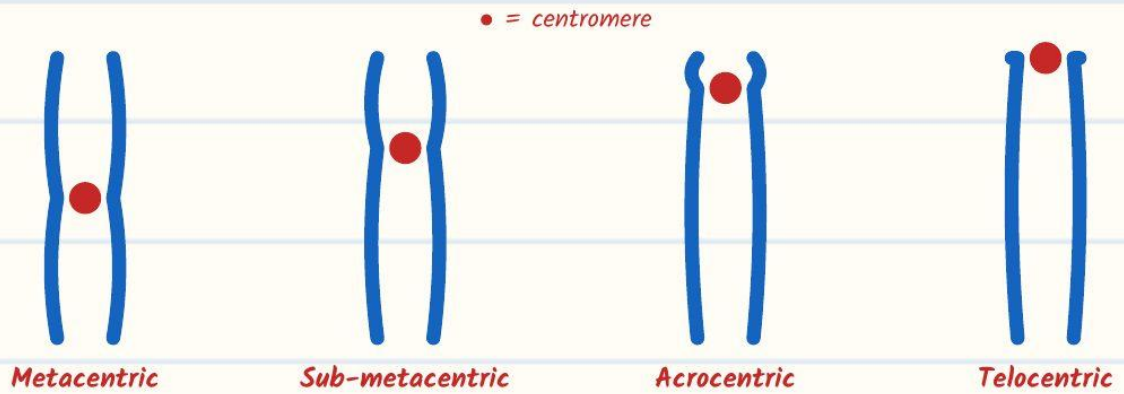


Axoneme: 9 + 2

- **Cytoskeleton**: microtubules, microfilaments, intermediate filaments → support, motility, shape
- Cilia small, many, oar-like · flagella longer, few
- Covered by plasma membrane · core = **axoneme**: 9 peripheral doublets + 2 central singlets = 9 + 2 array ★ exam
- Radial spokes join peripheral doublets to central sheath · arise from **basal body** (centriole-like)
- **Centrosome** = 2 perpendicular **centrioles** in pericentriolar material · each = 9 evenly spaced triplets of tubulin, cartwheel · central **hub** + radial spokes
- Centrioles form basal bodies of cilia/flagella and **spindle fibres** in animal cell division
- **Microbodies** — membrane-bound vesicles with enzymes, in plant & animal cells

8 Nucleus & Chromosomes

- Nucleus first described by **Robert Brown** (1831) · **chromatin** named by Flemming
- **Nuclear envelope** — 2 membranes + perinuclear space (10–50 nm) · outer membrane continuous with ER · **nuclear pores** let RNA & proteins pass
- Nucleoplasm: **nucleolus** (no membrane; active **rRNA synthesis**) + chromatin
- Chromatin = DNA + basic **histone** proteins + non-histone proteins + RNA → condenses into chromosomes in division
- Each chromosome has a **centromere** (primary constriction) with disc-shaped **kinetochores**
- **Satellite** — small fragment beyond a **secondary constriction** on some chromosomes
- Mature RBCs (mammals) & sieve tube cells lack a nucleus

**Quick revision ✓**

Virchow: *omnis cellula-e cellula* · smallest *Mycoplasma*, largest ostrich egg ·
 mesosomes · 70S vs 80S · fluid mosaic = Singer & Nicolson · cis → trans Golgi ·
 tonoplast · cristae · grana / stroma lamellae · 9+2 axoneme · centriole 9 triplets ·
 nucleolus = rRNA · metacentric → telocentric